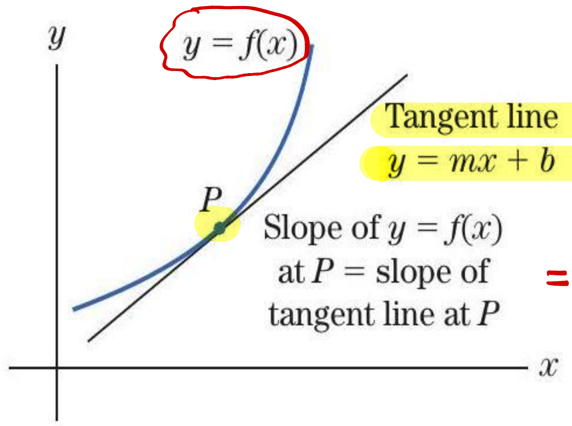
Chapter 4 Differentiation

The Slope of a Curve at a Point

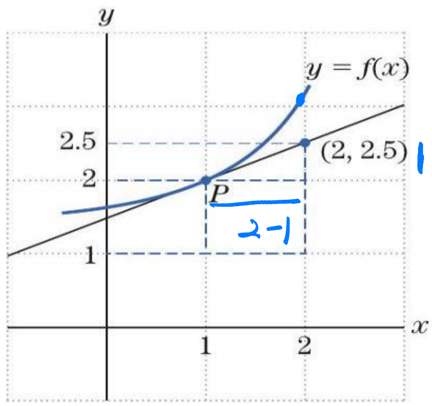
The slope of the graph at P is the slope of the tangent line at P.



$= m = ?$

Example 1

The graph of $y = f(x)$ and its tangent line at the point $P = (1,2)$ are shown. To find the slope of the graph at the point P, we note that the slope of the tangent line at P is $\frac{1}{2}$.



$2.5-2$

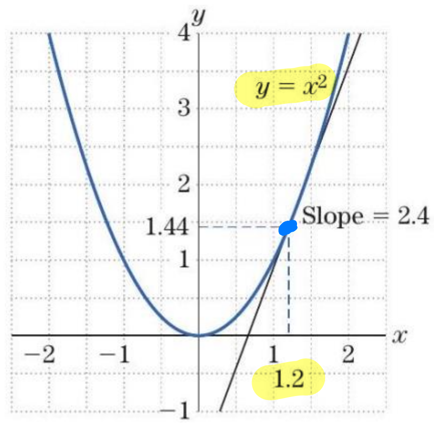
$m = \frac{2.5-2}{2-1} = \frac{0.5}{1} = \frac{1}{2}$

Since the slope of the graph is the slope of the tangent line, we conclude that the graph at P has slope $\frac{1}{2}$.

The slope of the tangent line is usually given by a formula. For example, the slope formula for the function $f(x) = x^2$ is $2x$.

Example 2

Consider the point $P = (1.2, 1.44)$ on the parabola $y = x^2$. To find the slope m of the tangent line to the parabola at P, we evaluate the slope formula $m = 2x$ at the first coordinate $x = 1.2$ of P and get $m = 2(1.2) = 2.4$. Thus, the tangent line at the point P has slope 2.4.



$y = x^2.$

$\text{slope} = m = 2x$

$x = 1.2$

$m = 2(1.2) = 2.4$

$m = \frac{dy}{dx}$

$m = \frac{d}{dx}(x^2)$

$m = 2x$

The Derivative and Limits

The derivative of $y = f(x)$ is the slope formula that gives the slope of the tangent line to the graph at a point (x, y) on the graph of $y = f(x)$. The derivative is denoted by $f'(x)$ or $\frac{d}{dx}[f(x)]$

$\frac{dy}{dx}$

Derivative formulas:

If $f(x) = mx + b$, then $f'(x) = m$.

(Derivative of a linear function is its slope.)

If $f(x) = c$, then $f'(x) = 0$.

(Derivative of a constant function is 0.)

x^n

$\boxed{\frac{d}{dx}}$

nx^{n-1}

$\frac{d}{dx}(x^n) = n x^{n-1}$

The Power Rule:

If $f(x) = x^n$, then $f'(x) = nx^{n-1}$.

Example 3

The following are examples that we obtain by applying the derivative formulas:

(a) $\frac{d}{dx}(-7x + 4) = -7$

$\frac{d}{dx}(-7x) + \frac{d}{dx}(4) = -7 + 0 = -7$

(b) $\frac{d}{dx}(3^2) = 0$ Derivative of a constant.

(c) $\frac{d}{dx}(x^3) = 3x^2$ Power rule with $n = 3$.

$\frac{d}{dx}(x^3) = 3x^{3-1} = 3x^2.$

(d) $\frac{d}{dx}\left[\frac{1}{\sqrt{x}}\right] = \frac{d}{dx}\left[x^{-1/2}\right]$

$$= -\frac{1}{2}x^{-\frac{1}{2}-1} \quad \text{Power rule with } n = -\frac{1}{2}$$

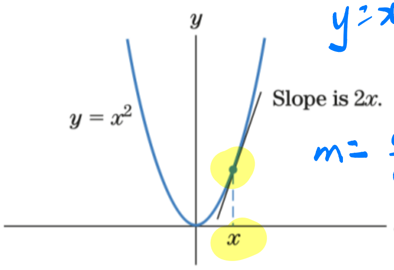
$$= -\frac{1}{2}x^{-\frac{3}{2}} = -\frac{1}{2x^{\frac{3}{2}}}$$

$\frac{d}{dx}(x^n) = n x^{n-1}$ $n = -1/2$.

$$\frac{d}{dx}(x^{-1/2}) = \left(-\tfrac{1}{2}\right) x^{\left(-\frac{1}{2}\right)-1} = \left(-\tfrac{1}{2}\right) x^{-\frac{3}{2}} \#$$

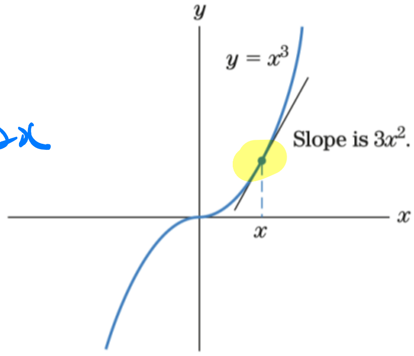
Geometric Meaning of the Derivative and Equation of the Tangent Line

We should, at this stage at least, keep the geometric meaning of the derivative clearly in mind.



$y = x^2$

$m = \frac{dy}{dx} = 2x$



$y = x^3$

$m = \frac{dy}{dx} = 3x^2$.

In general, the derivative of $f(x)$ at $x = a$ is the slope of the graph of $f(x)$ at the point $(a, f(a))$. Since the slope of the graph at a point is by definition the slope of the tangent line at that point, we have

Geometric Meaning of the Derivative

$$\begin{aligned} f'(a) &= \text{slope of the graph of } f(x) \text{ at } (a, f(a)) \\ &= \text{slope of the tangent line to the graph of } f(x) \text{ at } (a, f(a)) \end{aligned}$$

Example 4 Finding an Equation of the Tangent Line at a Given Point

To find the equation of the tangent line to the graph of $f(x) = x^5$ at the point $P = (-1, -1)$, we first find the derivative (slope formula) $f'(x) = 5x^4$. $= m$.

Evaluate the derivative at $x = -1$: $f'(-1) = 5(-1)^4 = 5$. $= m$

Thus, the slope of the tangent line at P is 5 and the point-slope equation of the tangent line is $y - (-1) = 5(x - (-1))$. $y - (-1) = m(x - (-1))$

$y + 1 = 5x + 5$ $5x - y + 4 = 0$ $ax + by + c = 0$, $a \geq 0$.

In general, to find the point-slope equation of the tangent line to the graph of $y = f(x)$ at the point with first coordinate $x = a$, proceed as follows:

Step 1 Find the point of contact of the graph and the tangent line by evaluating $f(x)$ at $x = a$. This yields the point $(a, f(a))$.

Step 2 Find the slope of the tangent line by evaluating the derivative $f'(x)$ at $x = a$. This yields the slope $m = f'(a)$.

Using the point $(a, f(a))$ and the slope $m = f'(a)$, we obtain the equation of the tangent line:

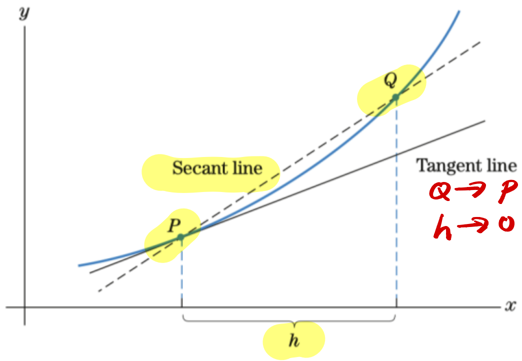
Equation of the Tangent Line

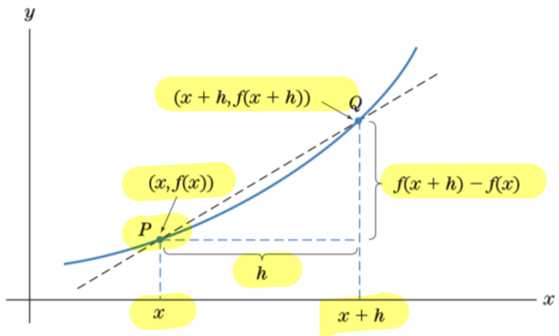
$$y - f(a) = f'(a)(x - a).$$

Limits and the Secant-Line Calculation of the Derivative

Suppose that the point P is $(x, f(x))$. Also, Q is h horizontal units away from P. Then, Q has x-coordinate $x + h$ and y-coordinate $f(x + h)$. The slope of the secant line through the points $P = (x, f(x))$ and $Q = (x + h, f(x + h))$ is simply

$$[\text{slope of secant line }] = \frac{f(x+h) - f(x)}{(x+h) - x} = \frac{f(x+h) - f(x)}{h}.$$





To move Q close to P along the curve, we let h approach zero. Then, the secant line approaches the tangent line, and so,
[slope of secant line] approaches [slope of tangent line];
that is,

$$h \to 0 \qquad \frac{f(x+h) - f(x)}{h} \text{ approaches } f'(x).$$

Let us summarize the process as follows:

Calculating the Derivative Informally

The three-step method

To calculate $f'(x)$:

1 Write the difference quotient $\frac{f(x+h)-f(x)}{h}$ for $h \neq 0$.

2 Simplify this difference quotient.

3 Let h approach zero. The quantity $\frac{f(x+h)-f(x)}{h}$ will approach $f'(x)$.

$h \to 0$

Another way of stating Step 3 is to say that the limit of $\frac{f(x+h)-f(x)}{h}$ is $f'(x)$, as h approaches 0 .

$\frac{d}{dx}(x^2) = 2x$?

Example 5

Verification of the Power Rule for $r = 2$. Apply the three-step method to show that the derivative of $f(x) = x^2$ is $f'(x) = 2x$.

1. $\frac{f(x+h)-f(x)}{h} = \frac{(x+h)^2 - x^2}{h}$, $h \neq 0$

2. $\frac{f(x+h)-f(x)}{h} = \frac{x^2+2xh+h^2-x^2}{h}$, $h \neq 0$

$= 2x+h$

3. $f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h} = \lim_{h\to 0} 2x+h$

$= 2x$

$f(x)=x^2$, $f'(x)=2x$.

Limits and the Derivative

Let $g(x)$ be a function and a a number on the x-axis that may or may not be in the domain of g. We say that the limit of $g(x)$ as x approaches a is L if the values of $g(x)$ can be made arbitrarily close to a number L by taking x sufficiently close (but not equal) to a. In symbols, we write

$$\lim_{x\to a} g(x) = L$$

Examples of limits:

(a) $\lim_{x\to 0} 2 = 2$

(b) $\lim_{x\to -1} x^3 = (-1)^3 = -1$

(c) $\lim_{x\to 4} (3x - 2) = 3(4) - 2 = 10$

(d) $\lim_{x\to 2} \frac{x^2-4}{x-2} = \lim_{x\to 2} \frac{\cancel{(x-2)}(x+2)}{\cancel{x-2}}$ Factor.

$\frac{0}{0}$

$x \neq 0$

$$= \lim_{x\to 2} (x + 2) = 4$$

(e) $\lim_{x\to 2}\frac{x^2+4}{x-2} \neq \frac{8}{0}$ Limit does not exist.

Limits arise naturally when we are studying derivatives. The limit definition of the derivative is

$$f'(x)=\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$$

When the limit $f'(x)$ exists, we say that f is differentiable at x.

If the difference quotient

$$\frac{f(x+h)-f(x)}{h}$$

does not approach any specific number as h approaches 0 , we say that f is nondifferentiable at x. Essentially, all the functions in this text are differentiable at all points in their domain.

Using Limits to Calculate $f'(a)$

Step 1 Write the difference quotient $\frac{f(a+h)-f(a)}{h}$.
Step 2 Simplify the difference quotient.
Step 3 Find the limit as $h\to 0$. This limit is $f'(a)$.

Example 6 Computing a Derivative from the Limit Definition

Use limits to compute the derivative $f'(5)$ for the following functions.
m
(a) $f(x)=15-x^2$
(b) $f(x)=\frac{1}{2x-3}$
(a)

1.
$$\begin{aligned}\frac{f(5+h)-f(5)}{h} &= \frac{[15-(5+h)^2]-(15-5^2)}{h} && \text{Step 1}\\ &= \frac{15-(25+10h+h^2)-(15-25)}{h} && \text{Step 2}\\ &= \frac{-10h-h^2}{h} = -10-h\end{aligned}$$

Therefore,

Step 3
$$f'(5)=\lim_{h\to 0}(-10-h)=-10.$$

(b)

step 1. $\frac{f(5+h)-f(5)}{h} = \frac{\frac{1}{2(5+h)-3}-\frac{1}{2(5)-3}}{h}$

$\vdots$

Step2. $= \frac{-2}{49+14h}$.

Step 3 $f'(5)=\lim_{h\to 0}\frac{-2}{49+14h} = -\frac{2}{49}$

$$\frac{f(5+h)-f(5)}{h}=\frac{\frac{1}{2(5+h)-3}-\frac{1}{2(5)-3}}{h}$$
$$=\frac{\frac{1}{7+2h}-\frac{1}{7}}{h}=\frac{\frac{7-(7+2h)}{(7+2h)7}}{h}$$
$$=\frac{-2h}{(7+2h)7\cdot h}=\frac{-2}{(7+2h)7}=\frac{-2}{49+14h}$$

step 2.

Therefore,

$$f'(5)=\lim_{h\to 0}\frac{-2}{49+14h}=-\frac{2}{49}.$$

Example 7 Computing Derivative from the Definition

Find the derivative of $f(x)=x^2+2x+2$.

$f'(x) = 2x+2$ ※ ?

Step 1. $\frac{f(x+h)-f(x)}{h} = \frac{(x+h)^2+2(x+h)+2-(x^2+2x+2)}{h}$

Step 2. $\vdots$

$= 2x+2+h$

Step 3

$f'(x) = \lim_{h\to 0}(2x+2+h)$

$= 2x+2$ ※

Some Rules for Differentiation

In computing derivatives, we can use basic rules for differentiation such as the following:

- Constant-multiple rule:

$$\frac{d}{dx}[k\cdot f(x)]=k\cdot\frac{d}{dx}[f(x)],\ k \text{ a constant.}$$

- Sum rule:

$$\frac{d}{dx}[f(x)+g(x)] = \frac{d}{dx}[f(x)] + \frac{d}{dx}[g(x)]$$

- General power rule:

$$\frac{d}{dx}([g(x)]^r) = r\cdot[g(x)]^{r-1}\cdot\frac{d}{dx}[g(x)]$$ Chain Rule.

Using properties of derivatives, compute the following derivatives:

(a) $\frac{d}{dx}[-7x^2] = -7\frac{d}{dx}[x^2]$ Constant-multiple rule.

$= (-7)2x = -14x.$ Power rule.

(b) $\frac{d}{dx}[\sqrt{x}-3x] = \frac{d}{dx}[\sqrt{x}] + \frac{d}{dx}[-3x]$ Sum rule.

$= \frac{d}{dx}[x^{1/2}] - 3\frac{d}{dx}[x]$ Constant-multiple rule.

$= \frac{1}{2}(x^{-1/2}) - 3 = \frac{1}{2\sqrt{x}} - 3.$

(c) $\frac{d}{dx}\left[\frac{1}{\sqrt{x^2+1}}\right] = \frac{d}{dx}\left[(x^2+1)^{-1/2}\right]$

$$= -\frac{1}{2}(x^2+1)^{-\frac{1}{2}-1}\frac{d}{dx}[(x^2+1)]$$
$$= -\frac{1}{2}(x^2+1)^{-\frac{3}{2}}(2x)$$
$$= -\frac{x}{(x^2+1)^{\frac{3}{2}}}.$$

EXAMPLE 8 Using the Constant-Multiple Rule

Calculate.

(a) $\frac{d}{dx}(2x^5)$

(b) $\frac{d}{dx}\left(\frac{x^3}{4}\right)$

(c) $\frac{d}{dx}\left(-\frac{3}{x}\right)$

(d) $\frac{d}{dx}(5\sqrt{x})$

SOLUTION

(a) With $k=2$ and $f(x)=x^5$, we have

$$\frac{d}{dx}(2\cdot x^5) = 2\cdot\frac{d}{dx}(x^5) = 2(5x^{\overset{5-1}{4}}) = 10x^4.$$ Constant-multiple rule.

(b) Write $\frac{x^3}{4}$ in the form $\frac{1}{4}\cdot x^3$. Then,

$$\frac{d}{dx}\left(\frac{x^3}{4}\right) = \frac{1}{4} \cdot \frac{d}{dx}(x^3) = \frac{1}{4}(3x^2) = \frac{3}{4}x^2.$$ Constant-multiple and power rules.

$$\frac{d}{dx}\left(\frac{x^3}{4}\right) = \frac{1}{4}\,\frac{d}{dx}(x^3)$$
$$= \frac{1}{4}(3)\,x^{3-1}$$
$$= \frac{3}{4}x^2 \quad ※$$

$$\frac{d}{dx}(x^n) = n x^{n-1}$$
$$\frac{d}{dx}(x^r) = r x^{r-1}$$

(c) $\frac{d}{dx}\left(-\frac{3}{x}\right)$
$= -3\,\frac{d}{dx}(x^{-1})$
$= -3\,(-1)\,x^{-1-1}$
$= 3x^{-2}$ ※
$= \frac{3}{x^2}$

(d) $\frac{d}{dx}(5\sqrt{x})$
$= 5\,\frac{d}{dx}(x^{\frac{1}{2}})$
$= 5\left(\frac{1}{2}\right)x^{\left(\frac{1}{2}-1\right)}$
$= \frac{5}{2}x^{-\frac{1}{2}}$ ※
$= \frac{5}{2\sqrt{x}}$

EXAMPLE 9 Using the Sum Rule

Find each of the following.

(a) $\frac{d}{dx}(x^3 + 5x)$
(b) $\frac{d}{dx}\left(x^4 - \frac{3}{x^2}\right)$

SOLUTION

(a) Let $f(x) = x^3$ and $g(x) = 5x$. Then,

$$\frac{d}{dx}(x^3 + 5x) = \frac{d}{dx}(x^3) + \frac{d}{dx}(5x) \qquad \text{Sum rule}$$
$$= 3x^2 + 5. \qquad \text{Power rule}$$

(b) The sum rule applies to differences as well as sums. Indeed, by the sum rule,

(b) $\frac{d}{dx}\left(x^4 - \frac{3}{x^2}\right) = \frac{d}{dx}(x^4) + \frac{d}{dx}\left(\frac{-3}{x^2}\right)$
$= 4x^{4-1} + (-3)\,\frac{d}{dx}(x^{-2})$
$= 4x^3 + (-3)(-2)\,x^{-2-1}$
$= 4x^3 + 6x^{-3}$ ※
$= 4x^3 + \frac{6}{x^3}$ ※

$$\frac{d}{dx}\left(x^4 - \frac{3}{x^2}\right) = 4x^3 + 6x^{-3}$$

EXAMPLE 10 Differentiating a Radical

Differentiate $\sqrt{1-x^2}$.
SOLUTION

$$\begin{aligned}\frac{d}{dx}\left(\sqrt{1-x^2}\right) &= \frac{d}{dx}\left[(1-x^2)^{1/2}\right] \\ &= \frac{1}{2}(1-x^2)^{\overset{\frac{1}{2}-1}{-1/2}} \cdot \frac{d}{dx}(1-x^2) \\ &= \frac{1}{2}(1-x^2)^{-1/2} \cdot (-2x) \\ &= \frac{-x}{(1-x^2)^{1/2}} = \frac{-x}{\sqrt{1-x^2}}\end{aligned}$$

EXAMPLE 11 Using the General Power Rule

Differentiate

$$y = \frac{1}{x^3+4x}. = (x^3+4x)^{-1}$$

SOLUTION

$$\begin{aligned}\frac{dy}{dx} &= (-1)(x^3+4x)^{-1-1}\frac{d}{dx}(x^3+4x) \\ &= -(x^3+4x)^{-2}(3x^2+4) \\ &= -\frac{3x^2+4}{(x^3+4x)^2}\end{aligned}$$

More about Derivatives

To denote the derivative of $y = f(x)$, we can use any one of the symbols y', $f'(x)$, or $\frac{d}{dx}[f(x)]$. The symbol $\frac{d}{dx}$ tells us to take the derivative with respect to x. If you want to take the derivative with respect to a variable other than x, then use that variable in the derivative symbol. For example, if you want to differentiate with respect to t, then use the symbol $\frac{d}{dt}$. In the presence of the symbol $\frac{d}{dt}$, we treat t as the independent variable and all other letters are treated as constants. So,

$$\frac{d}{dx}[x^3] = 3x^2, \text{ but } \frac{d}{dt}[x^3] = 0$$

$\uparrow$
as constant.

because the derivative of a constant (in this case, x^3) is 0.

Compute the given derivatives.

(a) $\frac{d}{dt}[x^2+3t-t^2]=\frac{d}{dt}[x^2]+3\frac{d}{dt}[t]-\frac{d}{dt}[t^2]$

constant

Sum rule and constant-multiple rule.

$=0+3-2t=-2t+3$

Treat x as a constant.

(b) $\frac{d}{dx}[xt+t^2-7x^3]=\frac{d}{dx}[xt]+\frac{d}{dx}[t^2]-7\frac{d}{dx}[x^3]$

$=t\frac{d}{dx}[x]+0-7(3x^2)=t-21x^2$

Treat t as a constant.

Second derivatives

When we differentiate a function $f(x)$, we obtain a new function $f'(x)$. Since $f'(x)$ is itself a function, we can take its derivative and obtain what is called the second derivative of $f(x)$, denoted by $f''(x)$, is defined by:

$$f''(x)=\frac{d}{dx}[f'(x)]$$

You can keep on differentiating the derivatives. For example, the third derivative is $f'''(x)=\frac{d}{dx}[f''(x)]$, and so on. Another convenient notation for the higher-order derivatives of $y=f(x)$ is illustrated by the following:

$$f''(x)=\frac{d^2}{dx^2}[f(x)];\ f'''(x)=\frac{d^3}{dx^3}[f(x)]$$

EXAMPLE 12 Evaluating a Second Derivative

If $y=x^4-5x^3+7$, find $\left.\frac{d^2y}{dx^2}\right|_{x=3}$.

SOLUTION

1. $$\frac{dy}{dx}=\frac{d}{dx}(x^4-5x^3+7)=4x^3-15x^2$$

2. $$\frac{d^2y}{dx^2}=\frac{d}{dx}(4x^3-15x^2)=12x^2-30x$$

3. $$\left.\frac{d^2y}{dx^2}\right|_{x=3}=12(3)^2-30(3)=108-90=18.$$

EXAMPLE 13 Evaluating First and Second Derivatives

If $s = t^3 - 2t^2 + 3t$, find $\left.\frac{ds}{dt}\right|_{t=-2}$ and $\left.\frac{d^2s}{dt^2}\right|_{t=-2}$.

SOLUTION

$$\frac{ds}{dt} = \frac{d}{dt}(t^3) - 2\frac{d}{dt}(t^2) + 3\frac{d}{dt}(t)$$
$$= 3t^2 - 2(2)t + 3(1)$$
$$= 3t^2 - 4t + 3$$

$$\left.\frac{ds}{dt}\right|_{t=-2} = 3(-2)^2 - 4(-2) + 3$$
$$= 23 \text{ ※}$$

$$\frac{d^2s}{dt^2} = \frac{d}{dt}(3t^2 - 4t + 3)$$
$$= 3(2)t - 4$$
$$= 6t - 4$$

$$\left.\frac{d^2s}{dt^2}\right|_{t=-2} = 6(-2) - 4$$
$$= -16 \text{ ※}$$

The Derivative as a Rate of Change

Average rate of change

Let $y = f(x)$ be a function defined on the interval $a \le x \le b$. Then,

$$\begin{bmatrix} \text{average rate of change of } f(x) \\ \text{over the interval } a \le x \le b \end{bmatrix} = \frac{f(b) - f(a)}{b - a}.$$

Example 14 The average rate of change

The average rate of change of $f(x) = \frac{1}{x}$ on the interval $\overset{a}{1} \le x \le \overset{b}{3}$ is

$$\frac{f(b) - f(a)}{b - a} = \frac{f(3) - f(1)}{3 - 1} = \frac{\frac{1}{3} - 1}{2} = \frac{-\frac{2}{3}}{2} = -\frac{2}{6} = -\frac{1}{3}$$

The (instantaneous) rate of change

The (instantaneous) rate of change of $f(x)$ at the point where $x = a$ is $f'(a)$.

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}.$$

$f(x) = x^{-1}$ $f'(x) = (-1)x^{-1-1}$
$= -x^{-2}$
$= -\frac{1}{x^2}$

Example 15

The instantaneous rate of change of $f(x) = \frac{1}{x}$ at $x = 3$ is $f'(3)$:

$$f'(x) = -\frac{1}{x^2};\ f'(3) = -\frac{1}{9}$$

Velocity and Acceleration

If $s(t)$ denotes the position function of an object moving in a straight line, then the velocity $v(t)$ of the object at time t is given by

$$v(t) = s'(t)$$

The derivative of the velocity function $v(t)$ is called the acceleration:

$$a(t) = v'(t) \text{ or } a(t) = s''(t)$$

Example 16

The position of a ball thrown up into the air is given by

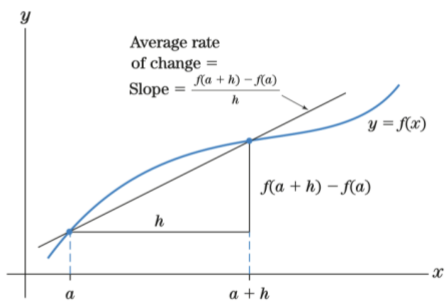
$$s(t) = -16t^2 + 64t + 5$$

where t is in seconds and $s(t)$ in feet. To find the time when the velocity is 0 , we first find the velocity at any time t, then set it equal to 0 and solve for t :

$$v(t) = s'(t) = -32t + 64$$
$$-32t + 64 = 0$$
$$t = \frac{64}{32} = 2$$

Thus, the velocity is 0 when $t = 2$ seconds.

Approximating Change with the Derivative



$f'(a) \approx \frac{f(a+h) - f(a)}{h}$

To approximate the change in a function, we can use the formula

- $f(a + h) - f(a) \approx f'(a) \cdot h$

If we let $a + h = x$, then $h = x - a$, and the formula becomes $f(x) - f(a) \approx f'(a) \cdot (x - a)$.

$f(x) - f(a) \approx f'(a) \cdot (x-a)$

Equivalently,

$$f(x) \approx f(a) + f'(a) \cdot (x - a)$$

$c(x) \approx c(a) + c'(a) \cdot (x-a)$

Example 17

Given $C(2) = 3.5$ and $C'(2) = .7$, we can approximate $C(1.8)$ as follows: Take $a = 2, x = 1.8$; then, $a=2$

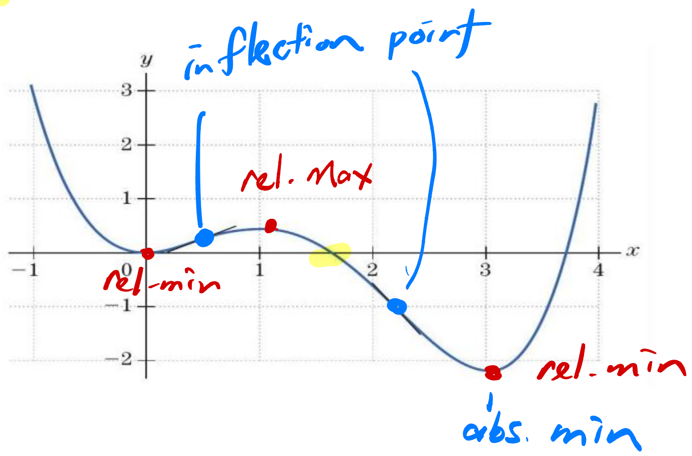
$C(x) \approx C(2) + C'(2)\,(x-2)$

$$\begin{aligned} C(1.8) &\approx C(2) + C'(2)(x - \overset{2}{a}) \\ &= 3.5 + (.7)(1.8 - 2) \\ &= 3.5 - .14 = 3.36. \end{aligned}$$

So, $C(1.8) \approx 3.36$.

Describing Graphs of Functions

1. Intervals in which the function is increasing (respectively decreasing), relative maximum points, relative minimum points
2. Maximum value, minimum value
3. Intervals in which the function is concave up (respectively, concave down), inflection points
4. x-intercepts, y-intercept
5. Undefined points
6. Asymptotes



EXAMPLE 18

Describe the graph in above, using the six categories listed previously.

1. Decreasing for $x < 0$ and $1 < x < 3$.

Increasing for $0 < x < 1$ and $x > 3$.

Relative minimum points: $(0,0)$ and $(3, -2.2)$.

Relative maximum point: $(1, 0.4)$.

2. Absolute minimum point: $(3, -2.2)$.

No absolute maximum.

3. Concave up for $x < .5$ and $x > 2.2$.

Concave down on $.5 < x < 2.2$.

Inflection points: $(.5, .2)$ and $(2.2, -1)$.

4. x-intercepts: $(0,1.6,3.7)$.

y-intercept: $(0,0)$.

5. No undefined points.

6. No asymptotes.

The First- and Second-Derivative Rules

If $f'(c) > 0$, then $f(x)$ is increasing at $x = c$.

If $f'(c) < 0$, then $f(x)$ is decreasing at $x = c$.

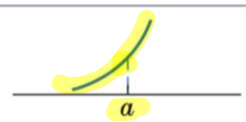
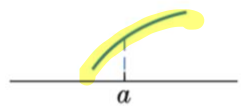
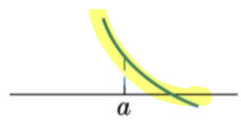
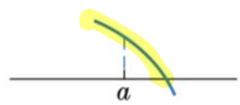
The second-derivative rule:

If $f''(c) > 0$, then $f(x)$ is concave up at $x = c$.

If $f''(c) < 0$, then $f(x)$ is concave down at $x = c$.

When $f''(a) = 0$, the second-derivative rule gives no information. In this case, the function might be concave up, concave down, or neither at $x = a$. «

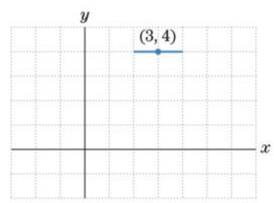
The following table shows how a graph may combine the properties of increasing, decreasing, concave up, and concave down.

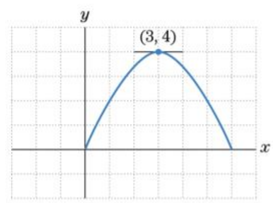
Conditions on the Derivatives	Description of $f(x)$ at $x = a$	Graph of $y = f(x)$ near $x = a$
1. $f'(a)$ positive $f''(a)$ positive	$f(x)$ increasing $f(x)$ concave up	
2. $f'(a)$ positive $f''(a)$ negative	$f(x)$ increasing $f(x)$ concave down	
3. $f'(a)$ negative $f''(a)$ positive	$f(x)$ decreasing $f(x)$ concave up	
4. $f'(a)$ negative $f''(a)$ negative	$f(x)$ decreasing $f(x)$ concave down	

EXAMPLE 19 Graphing Using Properties of the Derivative

Sketch the graph of a function $f(x)$ that has all the following properties.
(a) $f(3) = 4$ (3,4)
(b) $f'(x) > 0$ for $x < 3, f'(3) = 0$, and $f'(x) < 0$ for $x > 3$.

SOLUTION The only specific point on the graph is (3,4). We plot this point and then use the fact that $f'(3) = 0$ to sketch the tangent line at $x = 3$.
From property (b) and the first-derivative rule, we know that $f(x)$ must be increasing for x less than 3 and decreasing for x greater than 3.





$f''(3) < 0$
concave down.
(3,4) is a relative max point.

EXAMPLE 20 Graphing Using First and Second Derivatives [11]

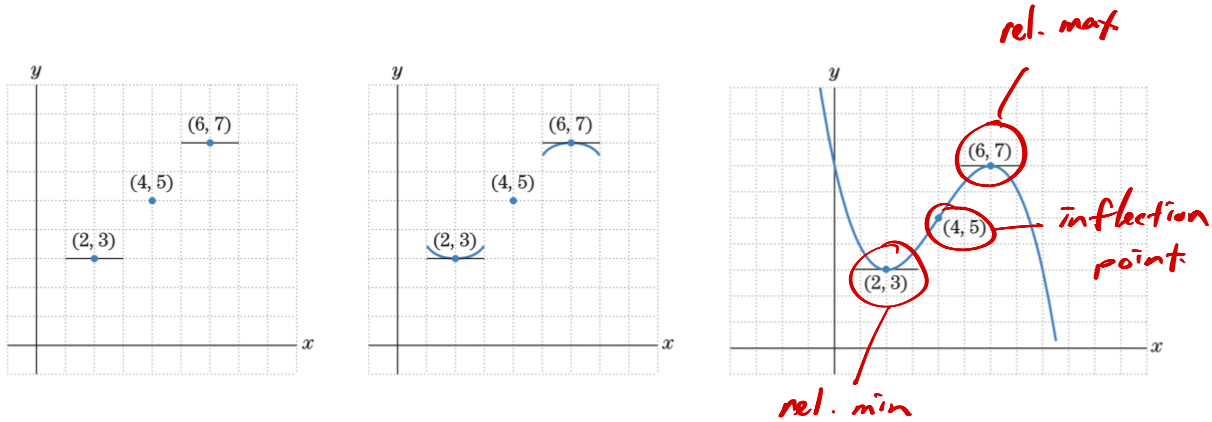
Sketch the graph of a function $f(x)$ with all the following properties.
(a) (2,3), (4,5), and (6,7) are on the graph.
(b) $f'(6) = 0$ and $f'(2) = 0$.

(c) $f''(x) > 0$ for $x < 4$, $f''(4) = 0$, and $f''(x) < 0$ for $x > 4$.

$f''(2) > 0$ concave up $\qquad$ $f''(6) < 0$ Concave down

SOLUTION $\quad$ (2,3) is relative min. $\qquad$ (6,7) is relative max

First, we plot the three points from property (a) and then sketch two tangent lines, using the information from property (b). From property (c) and the second derivative rule, we know that $f(x)$ is concave up for $x < 4$. In particular, $f(x)$ is concave up at (2,3). Also, $f(x)$ is concave down for $x > 4$, in particular, at (6,7). Note that $f(x)$ must have an inflection point at $x = 4$ because the concavity changes there. We now sketch small portions of the curve near (2,3) and (6,7). We can now complete the sketch taking care to make the curve concave up for $x < 4$ and concave down for $x > 4$.



The Second-Derivative Tests and Curve Sketching

The Second-Derivative Test (for local extreme points)

(a) If $f'(a) = 0$ and $f''(a) < 0$, then f has a relative local maximum at a.
(b) If $f'(a) = 0$ and $f''(a) > 0$, then f has a local minimum at a.

EXAMPLE 21 Applying the Second-Derivative Test [13]

The graph of the quadratic function $f(x) = \frac{1}{4}x^2 - x + 2$ is a parabola and so has one relative extreme point. Find it and sketch the graph.

SOLUTION We begin by computing the first and second derivatives of $f(x)$:

$$\begin{aligned} f(x) &= \frac{1}{4}x^2 - x + 2 \\ 1.\quad f'(x) &= \frac{1}{2}x - 1 = 0 \\ f''(x) &= \frac{1}{2}. \end{aligned}$$

Setting $f'(x) = 0$, we have $\frac{1}{2}x - 1 = 0$, so that $x = 2$ is the only critical value. Thus, $f'(2) = 0$.

$$y = f(2) = \frac{1}{4}(2)^2 - (2) + 2 = 1.$$

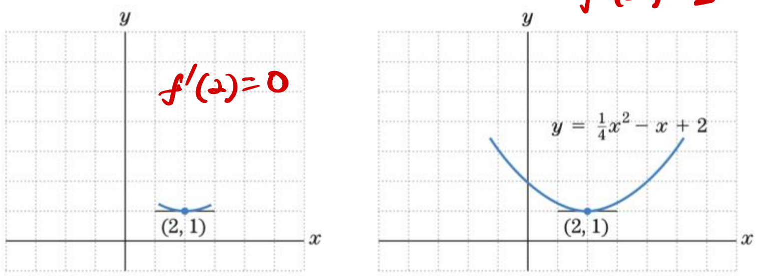
The critical point is (2,1).

$f''(2) = \frac{1}{2} > 0.$ U

point.

The graph of $f(x)$ is concave up at $x = 2$, so (2,1) is a local minimum by the second derivative test. A

$f''(2) = \frac{1}{2} > 0$



(2,1) is a min point.

EXAMPLE 22 Applying the Second-Derivative Test

Locate all possible relative extreme points on the graph of the function $f(x) = x^3 - 3x^2 + 5$. Check the concavity at these points and use this information to sketch the graph of $f(x)$:

SOLUTION We have

$$\begin{aligned} f(x) &= x^3 - 3x^2 + 5 \\ f'(x) &= 3x^2 - 6x \\ f''(x) &= 6x - 6. \end{aligned}$$

1. $$f'(x) = 3x^2 - 6x = 3x(x - 2) = 0$$

$x = 0$ and $x = 2$, we substitute these values back into the original expression for $f(x)$. That is, we compute

$$\begin{aligned} f(0) &= (0)^3 - 3(0)^2 + 5 = 5 \\ f(2) &= (2)^3 - 3(2)^2 + 5 = 1. \end{aligned}$$

The critical points are (0,5) and (2,1).

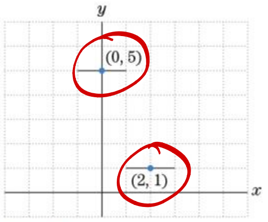


Figure 7

$f''(x) = 6x - 6$

$f''(x) = 0$
$6x - 6 = 0$
$x = 1$
$y = f(1) = 1 - 3 + 5$
$= 3$
$(1, 3)$

$x = 0$,	$f''(0) = 6(0) - 6 = -6 < 0$	Local maximum
$x = 2$	$f''(2) = 6(2) - 6 = 6 > 0.$	Local minimum

Since $f''(0)$ is negative, the graph is concave down at $x = 0$.

Since $f''(2)$ is positive, the graph is concave up at $x = 2$.

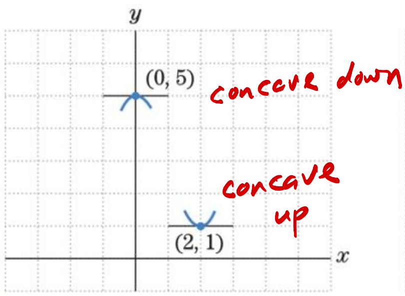


Figure 8

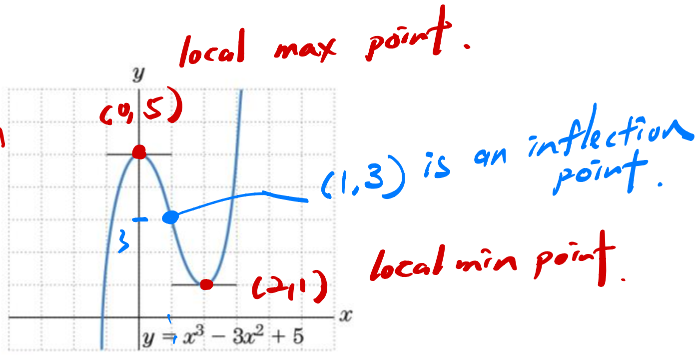


Figure 9

Locating Inflection Points

Under the assumption that $f''(x)$ is continuous, an inflection point of a function $f(x)$ can occur only at a value of x for which $f''(x)$ is zero, because the curve is concave up where $f''(x)$ is positive and concave down where $f''(x)$ is negative. Thus, we have the following rule:

Look for possible inflection points by setting

$$f''(x) = 0 \text{ and solving for } x.$$

Once we have a value of x where the second derivative is zero - say, at $x = b$,-we must check the concavity of $f(x)$ at nearby points to see if the concavity really changes at $x = b$.

EXAMPLE 23 Locating an Inflection Point

Find the inflection points of the function $f(x) = x^3 - 3x^2 + 5$.

SOLUTION

From Example 22, we have $f''(x) = 6x - 6 = 6(x - 1)$. $= 0$

$x = 1$

$y = f(1) = 3$

Clearly, $f''(x) = 0$ if and only if $x = 1$.

We will want to plot the corresponding point on the graph, so we compute

$$f(1) = (1)^3 - 3(1)^2 + 5 = 3.$$

Therefore, the only possible inflection point is $(1,3)$.

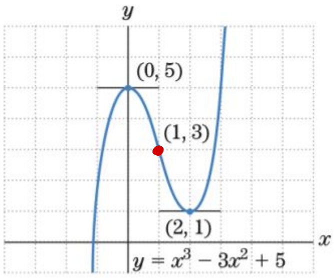


Figure 11

EXAMPLE 24 Graphing Using Properties of f' and f''

Sketch the graph of $y = -\frac{1}{3}x^3 + 3x^2 - 5x$ use second derivative test

SOLUTION

Let

$$f(x) = -\frac{1}{3}x^3 + 3x^2 - 5x.$$

Then,

$$f'(x) = -x^2 + 6x - 5$$
$$f''(x) = -2x + 6$$

Step 1 Finding the critical points

We set $f'(x) = 0$ and solve for x : $f'(x) = 0$

$$-(x^2 - 6x + 5) = 0$$
$$-(x-1)(x-5) = 0$$
$$x = 1 \text{ or } x = 5 \text{ Critical values}$$

Substituting these values of x back into $f(x)$, we find that

$$f(1) = -\frac{1}{3}(1)^3 + 3(1)^2 - 5(1) = -\frac{7}{3}$$
$$f(5) = -\frac{1}{3}(5)^3 + 3(5)^2 - 5(5) = \frac{25}{3}$$

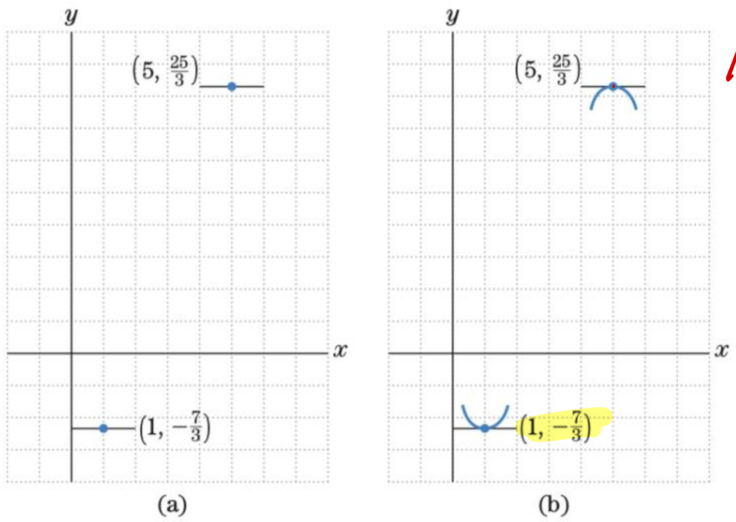
The critical points are $\left(1, -\frac{7}{3}\right)$ and $\left(5, \frac{25}{3}\right)$.

Step 2 Determining the extreme points

$f''(x) = -2x + 6$.

$x=1$: $\quad f''(1) = -2(1) + 6 = 4 > 0 \quad$ Local minimum

$x=5$: $\quad f''(5) = -2(5) + 6 = -4 < 0 \quad$ Local maximum



local max point

local min point

Figure 12

The curve is concave up at $x = 1$ because $f''(1)$ is positive (local minimum), and the curve is concave down at $x = 5$ because $f''(5)$ is negative (local maximum).

$\left(1, -\frac{7}{3}\right)$ is the local minimum point.

$\left(5, \frac{25}{3}\right)$ is the local maximum point.

Step 3 Concavity and inflection points

Since the concavity reverses somewhere between $x = 0$ and $x = 5$, there must be at least one inflection point. If we set $f''(x) = 0$, we find that

$$\begin{aligned} -2x + 6 &= 0 \\ x &= 3. \end{aligned}$$

So the inflection point must occur at $x = 3$. To plot the inflection point, we compute

$$f(3) = -\frac{1}{3}(3)^3 + 3(3)^2 - 5(3) = 3.$$

(3,3) is the inflection point.

The final sketch of the graph is given in Fig. 13.

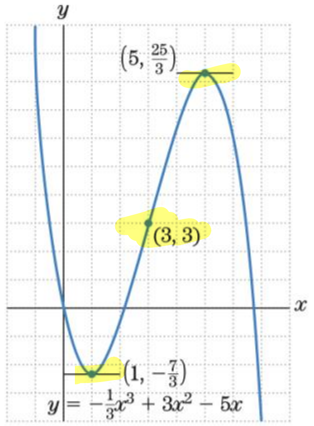


Figure 13

Local max point: $(5, \frac{25}{3})$

Local min point: $(1, -\frac{7}{3})$

Inflection point: $(3, 3)$

Optimization Problems

EXAMPLE 25 Finding Extreme Points

Find the minimum value of the function

$$f(x) = 2x^3 - 15x^2 + 24x + 19 \text{ for } x \geq 0.$$

SOLUTION We compute the derivatives:

$$f'(x) = 6x^2 - 30x + 24$$
$$f''(x) = 12x - 30$$

The x-coordinate of the minimum point satisfies the equation

$$\begin{aligned} f'(x) &= 0 \\ 6x^2 - 30x + 24 &= 0 \\ 6(x-4)(x-1) &= 0 \\ x &= 1,4 \text{ Critical values} \end{aligned}$$

The corresponding critical points on the curve are

$$(1, f(1)) = (1,30)$$
$$(4, f(4)) = (4,3).$$

Applying the second-derivative test, we see that

$x=1$: $f''(1) = 12 \cdot 1 - 30 = -18 < 0$ Maximum

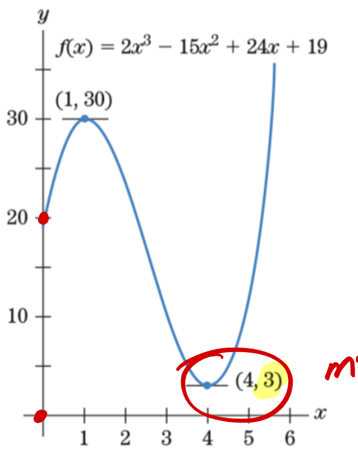
$x=4$: $f''(4) = 12 \cdot 4 - 30 = 18 > 0$ Minimum

That is, the point (1,30) is a relative maximum and the point (4,3) is a relative minimum.

At the left end point, $x = 0, f(0) = 19$, which is clearly larger than the relative minimum value 3.

Thus, the lowest point on the graph is $(4,3)$.

The minimum value of the function $f(x)$ is the y-coordinate of this point, 3.



min point.

$x \geq 0$

The Product and Quotient Rules

Product Rule

$$\frac{d}{dx}[f(x)g(x)] = f(x)g'(x) + g(x)f'(x)$$

For simplicity, we also write: $(fg)' = fg' + gf'$.

EXAMPLE 26 Finding the Derivative of a Product

Differentiate $y = (2x^3 - 5x)(3x + 1)$.

f g $f' = 6x^2 - 5$ $g' = 3$

SOLUTION

Let $f(x) = 2x^3 - 5x$ and $g(x) = 3x + 1$. Then,

g' f'

$$\begin{aligned}\frac{d}{dx}[(2x^3 - 5x)(3x + 1)] &= (2x^3 - 5x)\left(\frac{d}{dx}(3x + 1)\right) + (3x + 1)\cdot\left(\frac{d}{dx}(2x^3 - 5x)\right) \\ &= (2x^3 - 5x)(3) + (3x + 1)(6x^2 - 5) \\ &= 6x^3 - 15x + 18x^3 - 15x + 6x^2 - 5 \\ &= 24x^3 + 6x^2 - 30x - 5.\end{aligned}$$

simplify

EXAMPLE 27 Differentiating and Simplifying

Find

$$\frac{dy}{dx}, \quad \text{where } y = \underbrace{(x^2-1)^4}_{f}\underbrace{(x^2+1)^5}_{g}.$$

SOLUTION

$f' = 4(x^2-1)^3(2x) = 8x(x^2-1)^3$ ✓ $\quad g' = 5(x^2+1)^4(2x)$

$= 10x(x^2+1)^4$ ✓

$$\frac{dy}{dx} = fg' + gf'$$

$$= (x^2-1)^4\, 10x(x^2+1)^4 + (x^2+1)^5\, 8x(x^2-1)^3 \checkmark$$

$$= (2x)(x^2-1)^3(x^2+1)^4\left[5(x^2-1) + 4(x^2+1)\right]$$

simplify

$$= 2x(x^2-1)^3(x^2+1)^4\left[9x^2-1\right] \checkmark$$

The Quotient Rule

Another useful formula for differentiating functions is the quotient rule.

Quotient Rule

$$\frac{f}{g}$$

$$\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{g(x)f'(x) - f(x)g'(x)}{[g(x)]^2} = \frac{gf' - fg'}{g^2}$$

We must be careful to remember the order of the terms in this formula because of the minus sign in the numerator.

EXAMPLE 28 Using the Quotient Rule

Differentiate

$$y = \frac{x}{2x+3}. = \frac{f}{g}$$

$f' = \frac{d}{dx}(x) = 1$

$g' = \frac{d}{dx}(2x+3) = 2.$

SOLUTION Let $f(x) = x$ and $g(x) = 2x + 3$.

$$\frac{d}{dx}\left(\frac{x}{2x+3}\right) = \frac{(2x+3)\cdot \overbrace{\frac{d}{dx}(x)}^{f'} - (x)\cdot \overbrace{\frac{d}{dx}(2x+3)}^{g'}}{(2x+3)^2}$$

$$= \frac{(2x+3)\cdot 1 - x\cdot 2}{(2x+3)^2} = \frac{3}{(2x+3)^2}$$

$$\frac{gf' - fg'}{g^2}$$

EXAMPLE 29 Simplifying after the Quotient Rule

Find

$$\frac{dy}{dx}, \quad \text{where } y = \frac{x^3}{(x^2+1)^4}. = \frac{f}{g}$$

SOLUTION

$f(x) = x^3$ $\qquad g(x) = (x^2+1)^4$

$f' = 3x^2$ $\qquad g' = 4(x^2+1)^3(2x) = 8x(x^2+1)^3$

$$\frac{dy}{dx} = \frac{gf' - fg'}{g^2} = \frac{(x^2+1)^4\, 3x^2 - x^3\, 8x\, (x^2+1)^3}{(x^2+1)^8}$$

$$= \frac{(x^2+1)3x^2 - 8x^4}{(x^2+1)^5}$$

$$= \frac{x^2(3x^2+3-8x^2)}{(x^2+1)^5}$$

$$= \frac{x^2(3-5x^2)}{(x^2+1)^5}$$

simplify

#

The Chain Rule

A function of the form $[g(x)]^r$ is a composite $f(g(x))$, where the outside function is $f(x) = x^r$. We have already given a rule for differentiating this function:

$$\frac{d}{dx}[g(x)]^r = r[g(x)]^{r-1}g'(x).$$

The chain rule has the same form, except that the outside function $f(x)$ can be any differentiable function.

The Chain Rule

To differentiate $f(g(x))$, differentiate first the outside function $f(x)$ and substitute $g(x)$ for x in the result. Then, multiply by the derivative of the inside function $g(x)$. Symbolically,

$$\frac{d}{dx}f(g(x)) = f'(g(x))g'(x).$$

$$y = f(g(x)) = (x^5 + 9x + 3)^8$$

EXAMPLE 30 The General Power Rule as a Chain Rule

Use the chain rule to compute the derivative of $f(g(x))$, where $f(x) = x^8$ and $g(x) = x^5 + 9x + 3$.

$$\begin{aligned} f'(x) &= 8x^7,\ g'(x) = 5x^4 + 9 \\ f'(g(x)) &= 8(x^5 + 9x + 3)^7 \end{aligned}$$

SOLUTION

By the chain rule,

$$\frac{d}{dx}f(g(x)) = f'(g(x))g'(x) = 8(x^5 + 9x + 3)^7(5x^4 + 9).$$

$$\therefore \frac{d}{dx}\left((x^5 + 9x + 3)^8\right) = 8(x^5 + 9x + 3)^7(5x^4 + 9)$$ ※

EXAMPLE 31 The Chain Rule

Find $\frac{dy}{dx}$ if $y = u^5 - 2u^3 + 8$ and $u = x^2 + 1$.

$y = (x^2+1)^5 - 2(x^2+1)^3 + 8$

SOLUTION

$\frac{dy}{du} = 5u^4 - 2(3)u^2 + 0 = 5u^4 - 6u^2$

$\frac{du}{dx} = 2x$

$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$

$= (5u^4 - 6u^2)(2x)$

$= \left[5(x^2+1)^4 - 6(x^2+1)^2\right]2x$ ※

Differentiation of Exponential Functions

$$\frac{d}{dx}(e^x) = e^x.$$

Using this fact and the chain rule, we can differentiate functions of the form $y = e^{g(x)}$, where $g(x)$ is any differentiable function. The reason is that $e^{g(x)}$ is the composite of two functions. Indeed, if $f(x) = e^x$, then

$$e^{g(x)} = f(g(x)).$$

Thus, by the chain rule, we have

$$\begin{aligned}\frac{d}{dx}\left(e^{g(x)}\right) &= f'(g(x))g'(x)\\ &= f(g(x))g'(x) \text{ Since } f'(x) = f(x)\\ &= e^{g(x)}g'(x)\end{aligned}$$

So, we have the following result:

Chain Rule for Exponential Functions

Let $g(x)$ be any differentiable function. Then,

$$\frac{d}{dx}\left(e^{g(x)}\right) = e^{g(x)}g'(x)$$

If we write $u = g(x)$, then

$$\frac{d}{dx}(e^u) = e^u\frac{du}{dx}$$

EXAMPLE 32 Derivative of Functions of the Form $e^{g(x)}$

Differentiate

(a) $y = e^{5x}$ $= e^u$, $u = 5x$

(b) $y = e^{x^2-1}$ $= e^u$, $u = x^2 - 1$

(c) $y = e^{x-1/x}$ $= e^u$, $u = x - \frac{1}{x} = x - x^{-1}$ $\frac{dy}{dx} = 1 - (-1)x^{-2} = 1 + \frac{1}{x^2}$

SOLUTION Keep in mind that the derivative of $e^{g(x)}$ is $e^{g(x)}$ times $g'(x)$. So, in each part, identify $g(x)$, find $g'(x)$, and form the product $e^{g(x)}g'(x)$.

(a) Here $g(x) = 5x, g'(x) = 5$, so

$$\frac{d}{dx}(e^{5x}) = e^{5x}\frac{d}{dx}(5x) = e^{5x}\cdot 5 = 5e^{5x}.$$

(b) Here $g(x) = x^2 - 1, g'(x) = 2x$, so

$$\frac{d}{dx}\left(e^{x^2-1}\right) = e^{x^2-1}\frac{d}{dx}(x^2 - 1) = e^{x^2-1}\cdot(2x) = 2xe^{x^2-1}.$$

(c)

$y = e^{x - 1/x} = e^u$ $u = x - \frac{1}{x} = x - x^{-1}$

$\frac{dy}{dx} = 1 - (-1)x^{-2} = 1 + \frac{1}{x^2}$

$\frac{dy}{dx} = e^u \frac{du}{dx} = e^{x - 1/x}\left(1 + \frac{1}{x^2}\right)$ #

$y = \ln x$
$y = \log_e x.$
$e^y = x$

The Derivative of $\ln x$

Let us now compute the derivative of $y = \ln x$ for $x > 0$. Start with the equation

$$e^{\ln x} = x.$$

We will differentiate each side of this equation and then equate our results. Differentiate the left side, and get

$u = \ln x$

$$\frac{d}{dx}\left(e^{\ln x}\right) = e^{\ln x} \cdot \frac{d}{dx}(\ln x) \qquad \text{Use chain rule.}$$

$$= x \cdot \frac{d}{dx}(\ln x). \qquad \text{Use } e^{\ln x} = x.$$

$$= \frac{d}{dx}(x) = 1$$

Use chain rule.
Use $e^{\ln x} = x$.
Differentiate the right side, and get

$$\frac{d}{dx}(x) = 1.$$

$\frac{d}{dx}(e^x) = e^x$

Equating the two derivatives, we get

$$x \cdot \frac{d}{dx}(\ln x) = 1.$$

Dividing each side by $x > 0$, we obtain

Derivative of $y = \ln x$

$$\Rightarrow \boxed{\frac{d}{dx}(\ln x) = \frac{1}{x}}, \quad x > 0.$$

EXAMPLE 33 Derivatives Involving In x

Differentiate
(a) $y = (\ln x)^5$
(b) $y = x\ln x$
SOLUTION
(a) By the general power rule,

$$\frac{d}{dx}(\ln x)^5 = 5(\ln x)^{\overset{5-1}{4}} \cdot \frac{d}{dx}(\ln x) = 5(\ln x)^4 \cdot \frac{1}{x} = \frac{5(\ln x)^4}{x}.$$

(b) By the product rule,

$$\frac{d}{dx}(x\ln x) = x \cdot \frac{d}{dx}(\ln x) + (\ln x) \cdot \overset{\frac{d}{dx}(x)}{1} = \cancel{x} \cdot \frac{1}{\cancel{x}} + \ln x = 1 + \ln x.$$

Let $g(x)$ be any differentiable function. For any value of x for which $g(x)$ is positive, the function $y = \ln(g(x))$ is defined. For such a value of x, the derivative is given by the chain rule as

Chain Rule for Log Function

$$\frac{d}{dx}[\ln\ g(x)] = \frac{1}{g(x)} \cdot \frac{d}{dx} g(x) = \frac{g'(x)}{g(x)}.$$

If $u = g(x)$, equation (4) can be written in the form

$$\frac{d}{dx}[\ln\ u] = \frac{1}{u}\frac{du}{dx}.$$

EXAMPLE 34 Derivatives of Functions of the Form $y = \ln[g(x)]$

Differentiate

(a) $y = \ln(2x + 1)$ $= \ln(g(x))$

(b) $y = \ln(4x^2 - 2x + 9)$

(c) $y = \ln(xe^x)$

SOLUTION In each case we use (2).

(a) Here, $g(x) = 2x + 1$, $g'(x) = 2$, and so,

$$\frac{d}{dx}[\ln(2x + 1)] = \frac{1}{2x + 1}\frac{d}{dx}(2x + 1) = \frac{2}{2x + 1}.$$

(b) Here, $g(x) = 4x^2 - 2x + 9$, $g'(x) = 8x - 2$, and so,

$$\frac{d}{dx}[\ln(4x^2 - 2x + 9)] = \frac{1}{4x^2 - 2x + 9}\frac{d}{dx}(4x^2 - 2x + 9) = \frac{8x - 2}{4x^2 - 2x + 9}.$$

(c)

$y = \ln(xe^x) = \ln(u)$

$u = xe^x = fg$

$f(x) = x$

$g(x) = e^x$

$\frac{du}{dx} = fg' + gf'$

$= xe^x + e^x$

$= e^x(x+1)$

$f' = 1$

$g' = e^x$

$\frac{dy}{dx} = \frac{1}{u}\frac{du}{dx}$

$= \frac{1}{x\cancel{e^x}}(\cancel{e^x}(x+1))$

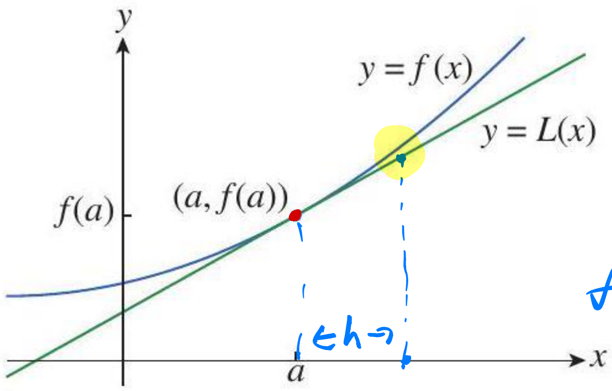
$= \frac{x+1}{x}$

$\frac{dy}{dx} = 1 + \frac{1}{x}$ ※

Linear Approximations and Differentials

Suppose the function f is differentiable at $x = a$ and the value $f(a)$ is known. Let L be

$h = x - a, \quad x = a + h.$



$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$

$f'(a) \approx \frac{f(x) - f(a)}{h}$

$h \to 0$

It seems very reasonable, and the graph suggests, that we can use the line tangent to the graph of f at $(a, f(a))$ as an approximation to the graph $y = f(x)$ when x is near a. An equation of the tangent line is

$$y = f(a) + f'(a)(x - a)$$

and the approximation

$$f(x) \approx f(a) + f'(a)(x - a) \tag{1}$$

high cost low cost

EXAMPLE 35 Linear Approximation

Let $f(x) = \sqrt{x+3}$. Find an equation of the tangent line to the graph of f at the point where $x = 1$ and use this linearization to approximate $f(1.05)$ and $f(0.98)$.

Solution high cost $a = 1$

$f(x) = \sqrt{x+3} = (x+3)^{1/2}$ Rewrite $f(x)$ using an exponent.

Find the derivative of f: $f'(x) = \frac{1}{2}(x+3)^{-1/2} = \frac{1}{2\sqrt{x+3}}$.

The tangent line passes through $(1, f(1)) = (1, \sqrt{3+1}) = (1,2)$

and has slope $f'(1) = \frac{1}{2\sqrt{1+3}} = \frac{1}{4}$.

An equation of the tangent line is

$$y = f(1) + f'(1)(x-1) = 2 + \frac{1}{4}(x-1) = \frac{7}{4} + \frac{x}{4}.$$

and the linearization is (the equation of this tangent line)

$$L(x) = f(1) + f'(1)(x-1) = 2 + \frac{1}{4}(x-1).$$

$f(x) = \underline{(x+3)}^{1/2} \approx \underline{2 + \frac{1}{4}(x-1)} = L(x)$

When x is near 1, $f(x) \approx L(x)$. See Figure 3.63.

Therefore, $x=1.05$ without using calculator.

$$f(1.05) = \sqrt{1.05+3} = \sqrt{4.05} \approx L(1.05) = 2 + \frac{1}{4}(1.05-1) = 2.013$$

$$f(0.98) = \sqrt{0.98+3} = \sqrt{3.98} \approx L(0.98) = 2 + \frac{1}{4}(0.98-1) = 1.995$$

Actual value.

Calculator.

$\sqrt{4.05} \approx 2.01246$

$\sqrt{3.98} \approx 1.99499$

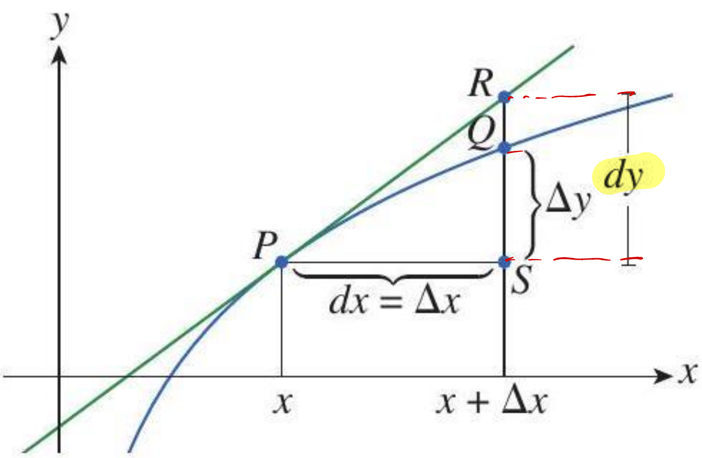
Differentials or small increment

The concept of a linear approximation is often developed using the terminology and notation associated with differentials. If $y = f(x)$, where f is a differentiable function, then the differential dx is considered an independent variable; that is, dx can take on the value of any real number. The differential $\boldsymbol{dy}$ is then defined in terms of dx by the equation

$$dy = f'(x)dx \tag{3}$$

Therefore, dy is a dependent variable; it depends on the values of x and dx. If dx is given a specific value and x is taken to be some specific number in the domain of f, then the numerical value of dy is determined (using Equation 3).

Here is a geometric interpretation of differentials.



$\Delta y \approx dy$

Let $P(x, f(x))$ and $Q(x + \Delta x, f(x + \Delta x))$ be points on the graph of f and let $dx = \Delta x$. The corresponding change in y is

$$\Delta y = f(x + \Delta x) - f(x)$$

The tangent line to the graph of $y = f(x)$ at the point P and through the point R has slope $f'(x)$. This slope must be equal to rise over run, or $\frac{dy}{dx}$. Therefore,

$f'(x) = \frac{dy}{dx} \Rightarrow f'(x)dx = dy.$

The differential dy represents the amount that the tangent lines rises or falls (or the change in the linearization). However, Δy represents the amount that the curve $y = f(x)$ rises or falls when x changes from x by an amount dx. Notice that the approximation $\Delta y \approx dy$ becomes better as Δx becomes smaller.

Suppose we let $dx = x - a$, then $x = a + dx$ and we can rewrite the linear approximation (Equation 1) using differential notation:

$$f(a + dx) \approx f(a) + dy$$

$(x+3)^{\frac{1}{2}}$

For example, for the function $f(x) = \sqrt{x+3}$ in Example 35, we have

$\Delta y \approx dy$

$$dy = f'(x)dx = \frac{dx}{2\sqrt{x+3}}$$

If $a = 1$ and $dx = \Delta x = 0.05$, then

$$dy = \frac{0.05}{2\sqrt{1+3}} = 0.013$$
$$\sqrt{4.05} = f(1.05) \approx f(1) + dy = 2.013$$

$$\begin{aligned}\Delta y &= f(1.05) - f(1)\\ &= (1.05+3)^{\frac{1}{2}} - (1+3)^{\frac{1}{2}}\\ &= (4.05)^{\frac{1}{2}} - 2\\ &= 0.01246 \approx 0.013 = dy\end{aligned}$$

Here is one more example that illustrates the use of differentials to estimate the errors that occur due to approximate measurements.

EXAMPLE 36 Error in Approximating the Volume of a Sphere

The radius of a sphere was measured and found to be 21 cm with a possible error in measurement of at most 0.05 cm. What is the maximum error in using this value of the radius to compute the volume of the sphere?

Solution

If the radius of the sphere is r, then the volume is $V(r) = \frac{4}{3}\pi r^3$.

If the error in the measured value of r is denoted by $dr = \Delta r$, then the corresponding error in the calculated value of V is ΔV. This value can be approximated by

$dV = V'(r)dr = 4\pi r^2 dr.$

For $r = 21, dr = 0.05$: $dV = 4\pi(21)^2(0.05) \approx 277.$

The maximum error in the calculated volume is approximately 277 cm^3.

Note: Often, a better way to describe this kind of inaccuracy is by using the relative error. This is computed by dividing the error (estimate) by the total volume:

$$\frac{\Delta V}{V} \approx \frac{dV}{V} = \frac{4\pi r^2 dr}{\frac{4}{3}\pi r^3} = 3\frac{dr}{r}$$

Therefore, in Example 36, the relative error in the volume is about three times the relative error in the radius.

Relative error in the radius is approximately $\frac{dr}{r} = \frac{0.05}{21} \approx 0.0024$. = 0.24%

Relative error in the volume is approximately $3(0.024) \approx 0.0072$. = 0.72%

These errors could also be expressed as percentage errors, for example, a 0.24% in the radius and 0.7% in the volume.

$y = f(x)$ - normal curve

Find the slope $\frac{dy}{dx}$ of the tangent line at a point on the graph defined by parametric equations

$y = f(t)$ $t \in \mathbb{R}$
$x = g(t)$.

$$\frac{dy}{dt} = \frac{dy}{dx}\frac{dx}{dt}$$

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} \text{ if } \frac{dx}{dt} \neq 0$$

EXAMPLE 37

Find an equation of the tangent line to the curve at the point corresponding to the given value of the parameter.

$x = t^4 + 1, y = t^3 + t;\ t = -1$

Solution

$x = t^4 + 1, y = t^3 + t; t = -1.$

$\frac{dy}{dt} = 3t^2 + 1, \frac{dx}{dt} = 4t^3$, and $\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{3t^2+1}{4t^3}$.

When $t = -1$, $(x, y) = (2, -2)$ and $\frac{dy}{dx} = \frac{4}{-4} = -1$, $= m$

so an equation of the tangent to the curve at the point corresponding to $t = -1$ is

$y - (-2) = (-1)(x - 2)$,

or $y = -x$.

$y - (-2) = m\,(x - 2)$

$y + 2 = (-1)(x - 2)$

$y + 2 = -x + 2$

$y = -x$

$x + y = 0$ #